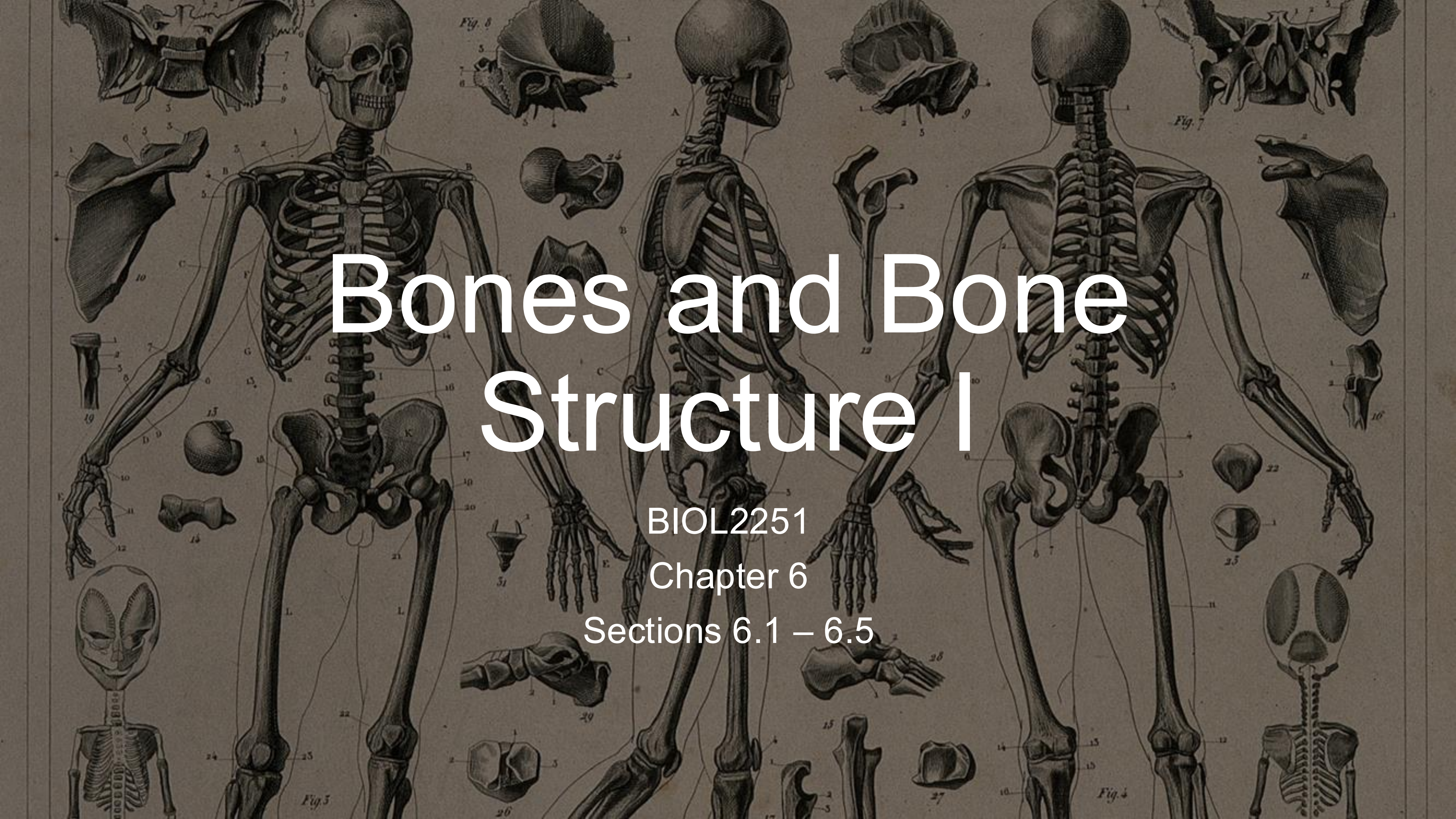


Bones and Bone Structure I

BIOL2251

Chapter 6

Sections 6.1 – 6.5

Functions of the Skeletal System

Functions of the Skeletal System

- **Instructions**
  - In one minute, **list as many functions of the skeletal system** as you can
- **Bloom's taxonomy = Remember/Understand**

Functions of the Skeletal System

Model Answer

- Support (framework for body)
- Protection (e.g., skull, ribcage)
- Movement (leverage with muscles)
- Mineral storage (calcium, phosphate)
- Blood cell production (red marrow)
- Energy storage (yellow marrow)

Bone Classification

Bone Classification

- **Instructions**
  - Working in pairs, **classify each of these bones according to their shape**
    - **Femur**
    - **Sternum**
    - **Vertebra**
    - **Patella**
    - **Frontal**
    - **Carpals**
- **Bloom’s taxonomy = Understand**

Bone Classification

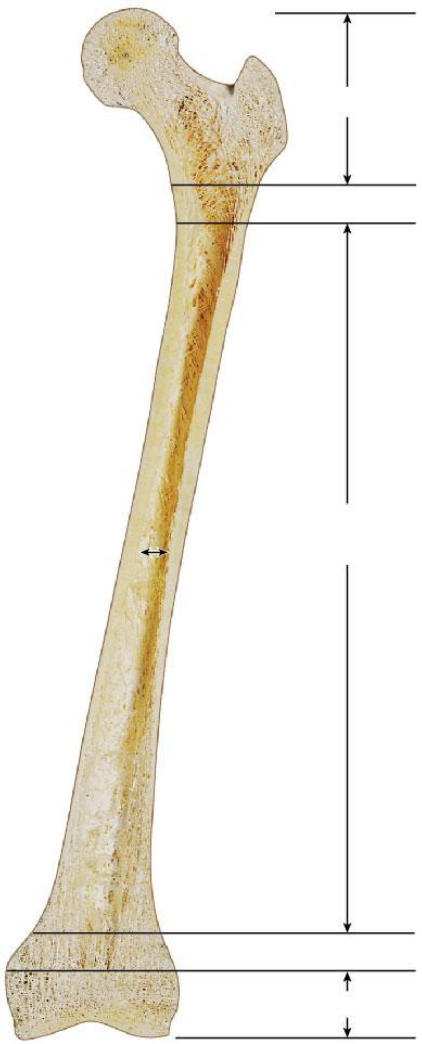
Model Answer

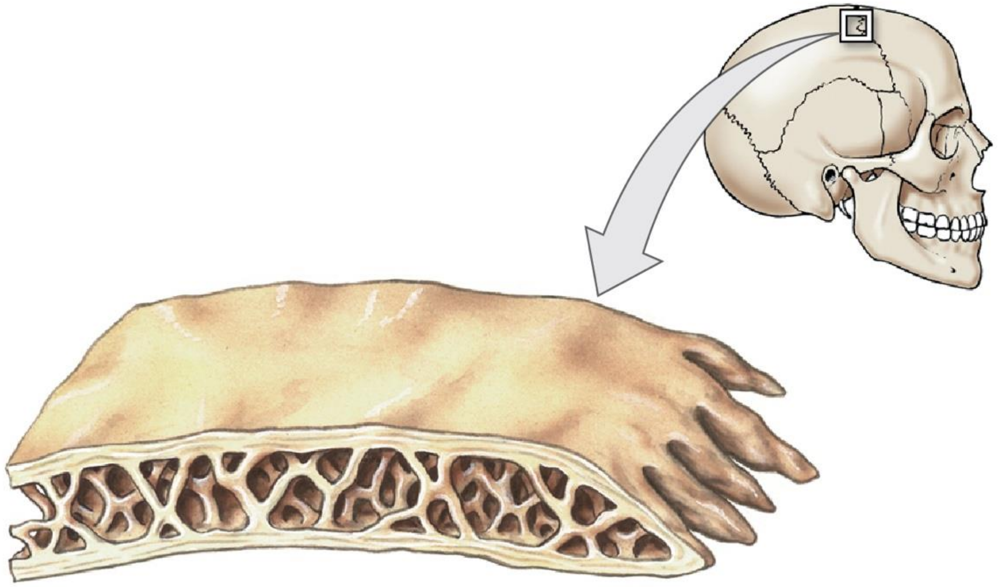
- Femur → Long
- Sternum → Flat
- Vertebra → Irregular
- Patella → Sesamoid
- Frontal bone → Flat
- Carpals → Short

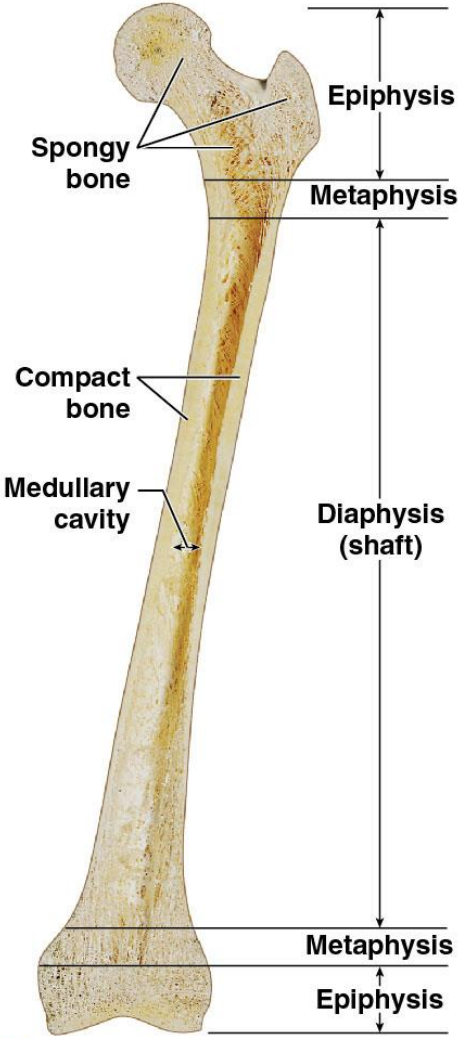
Long vs. Flat Bone Structures

Long vs. Flat Bone Structures

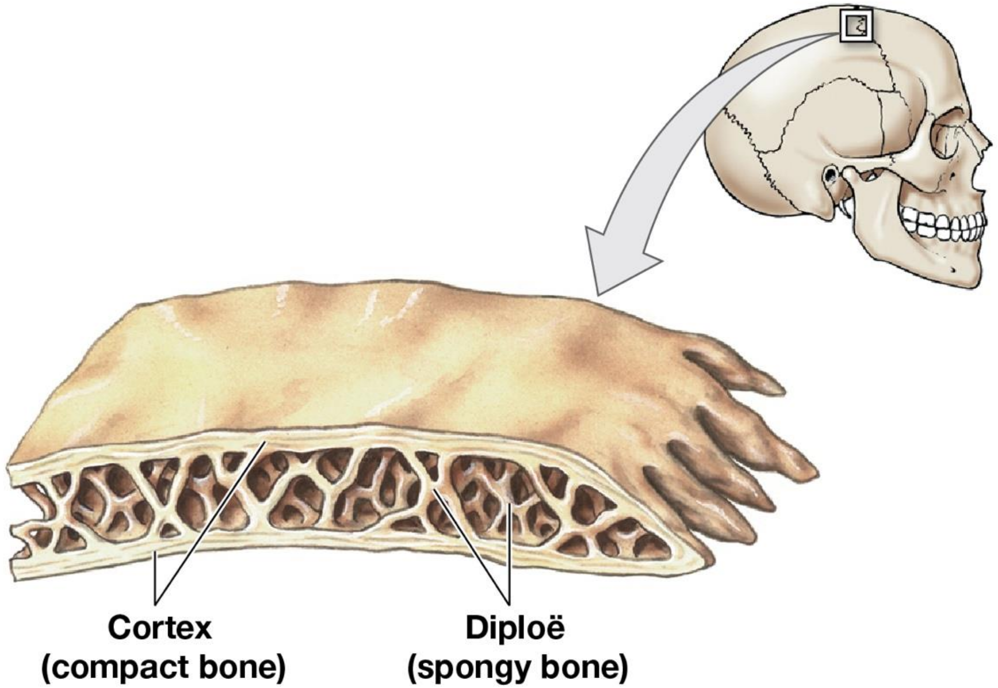
- **Instructions**
  - Form pairs or small groups
  - **Label the following structure on a long bone and a flat bone**
    - **Long bone:** diaphysis, epiphysis, metaphysis, medullary cavity, compact bone, spongy bone, periosteum
    - **Flat bone:** compact bone, spongy bone (diploë), periosteum
- **Bloom's taxonomy = Apply → Analyze**







a The structure of a representative long bone (the femur) in longitudinal section



b The structure of a flat bone (the parietal bone)

Bone Marking Vocabulary

Bone Marking Vocabulary

- **Instructions**
  - Work in pairs or in small groups
  - **Match each bone marking term to its correct description**
- **Bloom’s taxonomy = Remember → Apply**

Bone Marking Vocabulary Matching 1

- Condyle
- Foramen
- Crest
- Spine
- Trochanter
- Head
- Sinus

A. Rounded knob that articulates with another bone
B. Opening through bone for nerves/vessels
C. Hollow chamber within bone, usually air-filled
D. Prominent ridge
E. Pointed projection
F. Expanded end of a bone, supported by a narrower region
G. Large, rough projection

Bone Marking Vocabulary Matching 1

- Condyle = A. Rounded knob that articulates with another bone
- Foramen = B. Opening through bone for nerves/vessels
- Crest = D. Prominent ridge
- Spine = E. Pointed projection
- Trochanter = G. Large, rough projection
- Head = F. Expanded end of a bone, supported by a narrower region
- Sinus = C. Hollow chamber within bone, usually air-filled

Bone Marking Vocabulary Matching 2

- Fossa
- Tuberosity
- Facet
- Trochlea
- Neck
- Meatus
- Process

A. Smooth, flat surface for articulation

B. Shallow depression

C. Passage or channel

D. Narrow connection between head and shaft of bone

E. Any projection or bump

F. Pulley-shaped surface

G. Small projection for muscle attachment

Bone Marking Vocabulary Matching 2

- Fossa = B. Shallow depression
- Tuberosity = G. Small projection for muscle attachment
- Facet = A. Smooth, flat surface for articulation
- Trochlea = F. Pulley-shaped surface
- Neck = D. Narrow connection between head and shaft of bone
- Meatus = C. Passage or channel
- Process = E. Any projection or bump

Bone Matrix and Cells

Bone Matrix and Cells

- **Instructions**
  - Think on your own for a minute, then pair with a neighbor and discuss this question: **which bone cell is most important in repairing a fracture? Why?**
- **Bloom's taxonomy = Understand → Apply**

Bone Matrix and Cells

Model Answer

- **Osteoblasts** → produce new bone matrix to replace damaged tissue.
  - Supporting detail: osteo**clasts** clear debris, osteo**cytes** sense stress, osteo**genic** cells can become osteoblasts, but **osteoblasts are the builders**

Compact vs. Spongy Bone

Compact vs. Spongy Bone

- **Instructions**
  - Work in small groups
  - **Decide which type of bone predominates** in each scenario.
    - **Diaphysis of femur**
    - **Epiphysis of humerus**
- **Bloom's taxonomy = Analyze**

Compact vs. Spongy Bone

Model Answer

- **Diaphysis = compact bone** → withstands compression, supports body weight.
- **Epiphysis = spongy bone** → resists stress from many directions, lightens weight, houses marrow.

Periosteum vs Endosteum

Periosteum vs Endosteum

- **Instructions**
  - Work in pairs or small groups
  - Complete the table on the next slide to **compare periosteum and endosteum**
    - **Consider anatomy and physiology**
- **Bloom's taxonomy = Understand**

Periosteum vs Endosteum

Feature	Periosteum	Endosteum
Location		
Layers		
Composition		
Functions		
Growth Role		
Clinical Relevance		

Periosteum vs Endosteum

Feature	Periosteum	Endosteum
Location	**Outer surface** of bones (except at joints covered by articular cartilage)	**Inner surfaces** of bones: lining medullary cavity, central canals, trabeculae
Layers	Two layers: **(1) fibrous outer layer, (2) cellular/osteogenic inner** layer	Thin **cellular layer only**
Composition	**Dense irregular connective tissue + osteogenic cells**	**Osteoblasts, osteoclasts, osteogenic cells**
Functions	Anchors tendons and ligaments via perforating fibers; provides blood vessels and nerves; **essential for appositional growth and repair**	Bone growth; bone repair; regulates balance between osteoblast and osteoclast activity
Growth Role	**Appositional growth** (increase in diameter/thickness)	**Remodeling** from the inside; **turnover of marrow cavity and trabeculae**
Clinical Relevance	**Very sensitive to pain**; contributes to pain in fractures	Thinner, not pain-sensitive, but **critical in healing/remodeling**

Bone Growth and Cell Roles

Bone Growth and Cell Roles

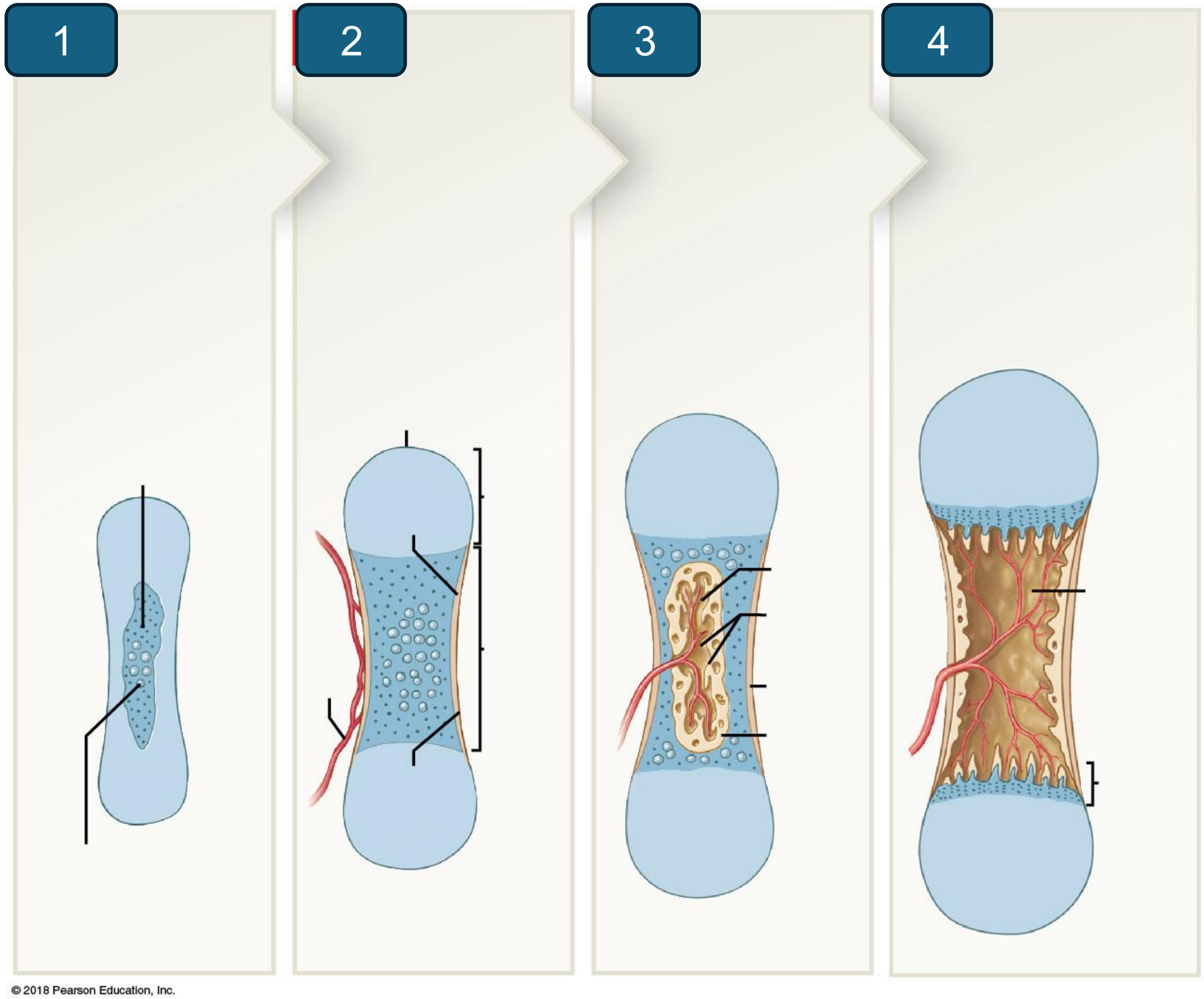
- **Instructions**
  - Let's **put together the process of endochondral ossification**
  - I will put the initial step on the board, then ask for volunteers to help us put the story together
- **Bloom's taxonomy = Analyze**

Bone Growth and Cell Roles

Cell Roles

- **Osteoblasts** → deposit new bone at surface.
- **Osteoclasts** → resorb bone to reshape medullary cavity, prevent bone from becoming too heavy.
- **Osteocytes** → embedded in matrix, sense stress and signal remodeling.

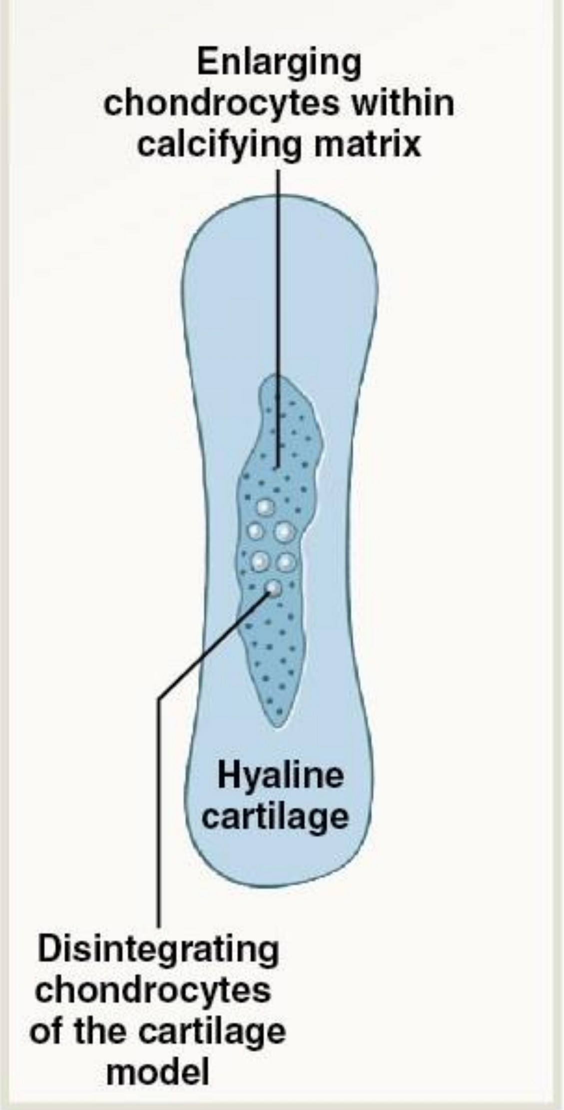
Endochondral Ossification



Endochondral Ossification: Step 1

Cartilage Model Formation

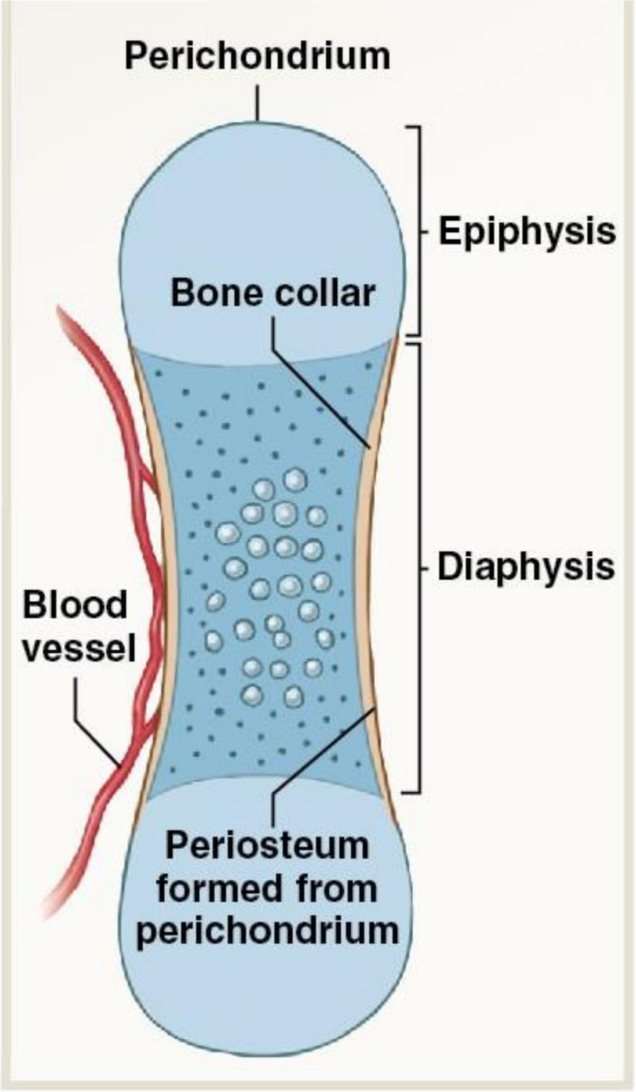
- Cartilage enlarges
- Chondrocytes increase in size
- Matrix begins to calcify
- Enlarged chondrocytes die, leaving cavities in the cartilage



Endochondral Ossification: Step 2

Bone Collar Formation

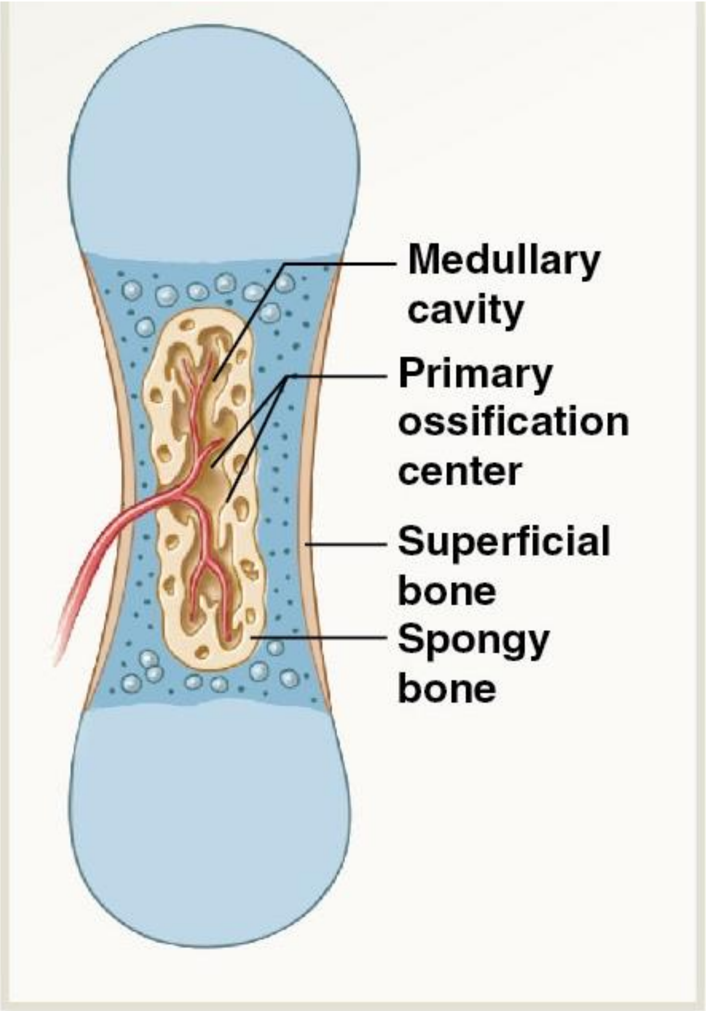
- Blood vessels enter the perichondrium
- Cells of perichondrium convert to osteoblasts
- Osteoblasts form a thin layer of bone around the shaft
- Perichondrium becomes periosteum



Endochondral Ossification: Step 3

Development of Primary Ossification Center

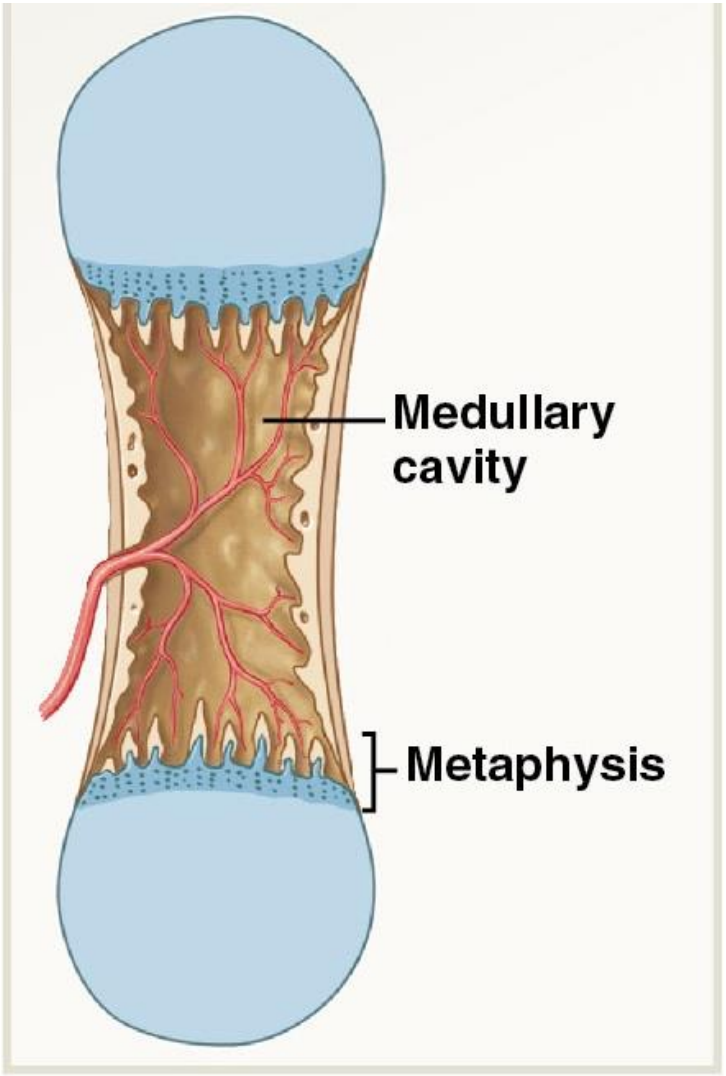
- Blood supply penetrates the center of the cartilage
- Osteogenic cells in the blood convert to osteoblasts and produce spongy bone
- Forms the **primary ossification center**
- Bone growth continues towards both ends of the cartilage model



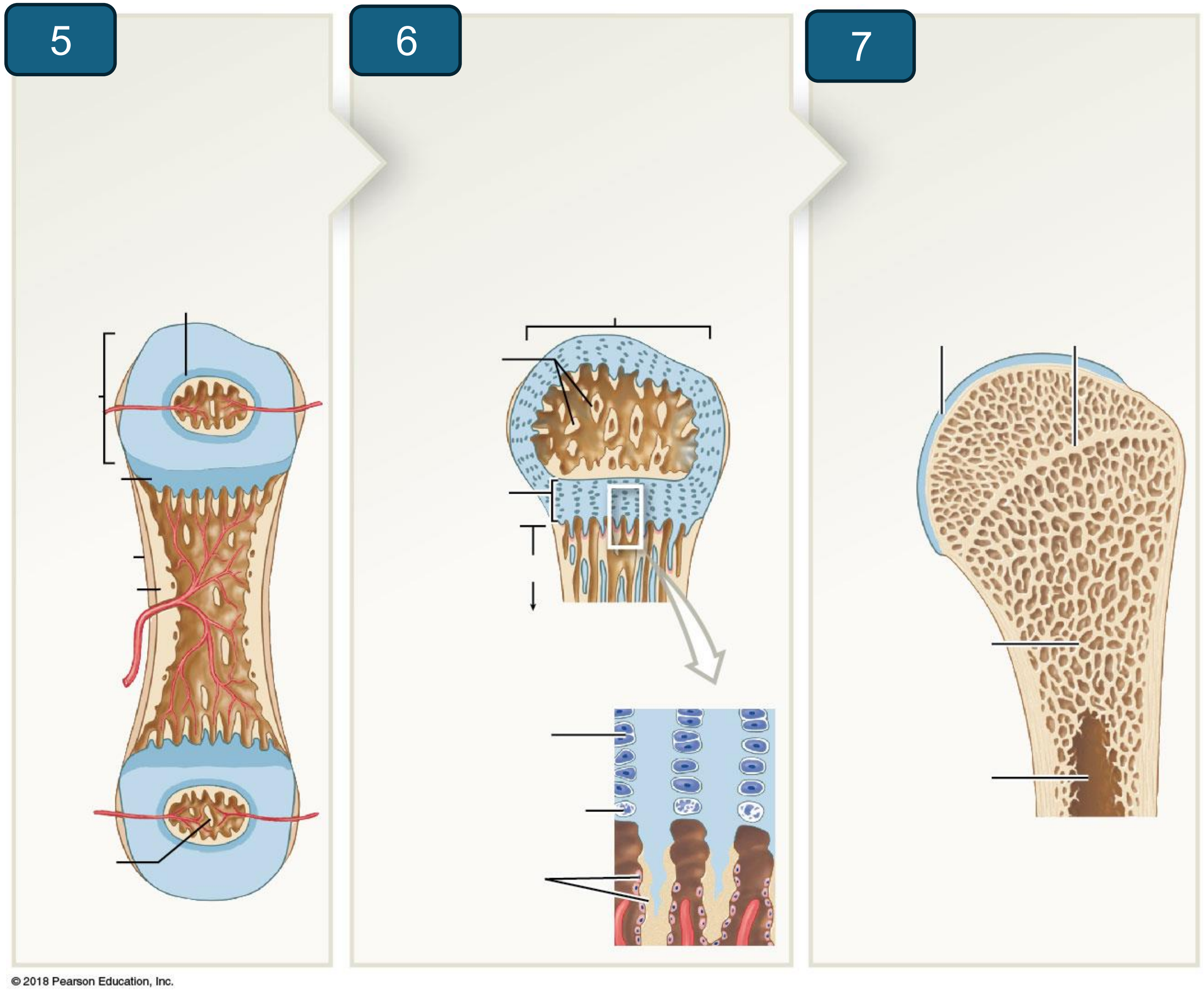
Endochondral Ossification: Step 4

Medullary Cavity Formation

- Osteoclasts erode the trabeculae of the spongy bone
- Medullary cavity is formed
- Diaphysis is fully formed
- Bone growth continues by increasing in *diameter* and *length* (Steps 5-7)
- **Appositional growth** = increase in *diameter* by osteoblasts in the periosteum



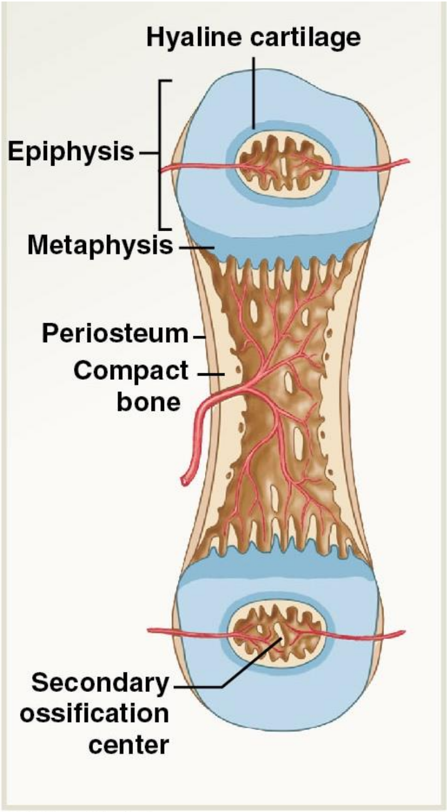
Endochondral Ossification



Endochondral Ossification: Step 5

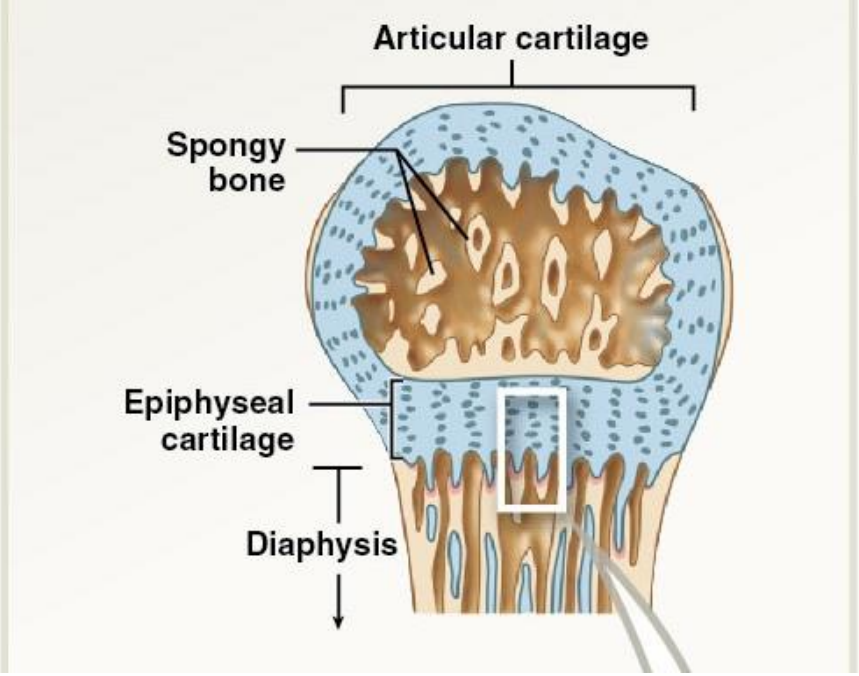
Development of Secondary Ossification Centers

- Capillaries and osteoblasts invade epiphyses
- Forms secondary ossification centers in the epiphyses (primary = diaphysis)
- Timing varies by bone and individual



Endochondral Ossification: Step 6

- Epiphyseal cartilage separates the epiphysis from diaphysis
- Epiphyses become filled with spongy bone
- Cap of original cartilage becomes **articular cartilage**
- Thin layer of cartilage at metaphysis **(epiphyseal cartilage)**
- **Chondrocytes create new cartilage at the same rate that osteoblasts make new bone**
  - If cartilage grows, bone will increase in length (**interstitial growth**)



Endochondral Ossification: Step 7

Epiphyseal Plate Closure (at puberty)

- **Osteoblasts work faster than chondrocytes**
- Bone growth is dramatically increased
- Epiphyseal cartilage is reduced until it is completely replaced by spongy bone **(epiphyseal closure)**
- Only remaining cartilage is the thin layer of hyaline **articular cartilage** that will reduce friction at synovial joints